

I. LAGRANGIAN FORMULATION

The dynamics of the inverted pendulum are derived using the Euler–Lagrange formulation. The Lagrangian is defined as

$$\mathcal{L} = T - V, \quad (1)$$

where T is the total kinetic energy and V is the gravitational potential energy.

For a system with generalized coordinates q_i , the Euler–Lagrange equations are

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \quad (2)$$

where Q_i is the generalized nonconservative force associated with q_i .

A. Coordinate System

The system is described using an inertial Cartesian coordinate frame $\mathcal{F}_I$. The x axis is horizontal and positive to the right, while the z axis is vertical and positive upward.

The position of the wheel axle is

$$\mathbf{r}_w = \begin{bmatrix} x \\ z \end{bmatrix}, \quad (3)$$

where z is constant.

The center of mass of the pendulum body is located at

$$\mathbf{r}_b = \begin{bmatrix} x_b \\ z_b \end{bmatrix} = \begin{bmatrix} x + \ell \sin \theta \\ z + \ell \cos \theta \end{bmatrix}, \quad (4)$$

where ℓ is the distance from the wheel axle to the center of mass of the pendulum body.

The angle θ is measured clockwise from the positive z axis. Therefore,

$$\theta = 0 \quad (5)$$

corresponds to the upright configuration of the inverted pendulum.

The generalized coordinates are

$$q_1 = x, \quad q_2 = \theta, \quad q_3 = \phi_w, \quad (6)$$

where ϕ_w is the angular position of the wheel.

The wheel angular velocity is

$$\omega_w = \dot{\phi}_w. \quad (7)$$

B. Rolling Constraint

Assuming pure rolling without slipping, the wheel translational and rotational velocities satisfy

$$\dot{x} = r_w \dot{\phi}_w, \quad (8)$$

where r_w is the wheel radius.

The corresponding position constraint is

$$x - r_w \phi_w = C, \quad (9)$$

where C is a constant determined by the initial configuration.

Because the rolling constraint eliminates one degree of freedom, ϕ_w may be eliminated from the final equations of motion using

$$\dot{\phi}_w = \frac{\dot{x}}{r_w}, \quad \ddot{\phi}_w = \frac{\ddot{x}}{r_w}. \quad (10)$$

C. Kinetic Energy

The translational kinetic energy of the wheel and pendulum body can first be represented using the combined mass and the horizontal velocity of the wheel axis. Since z is constant,

$$\dot{z} = 0.$$

The translational kinetic energy is therefore

$$T_{\text{trans}} = \frac{1}{2}(m_b + m_w)\dot{x}^2, \quad (11)$$

where m_b and m_w are the masses of the pendulum body and wheel assembly, respectively.

This expression does not account for the additional translational motion of the pendulum body center of mass caused by its angular displacement. The corresponding correction term is

$$T_{\text{corr}} = m_b \ell \dot{x} \dot{\theta} \cos \theta + \frac{1}{2} m_b \ell^2 \dot{\theta}^2. \quad (12)$$

The rotational kinetic energy of the pendulum body is

$$T_{b, \text{rot}} = \frac{1}{2} I_b \dot{\theta}^2, \quad (13)$$

and the rotational kinetic energy of the wheel is

$$T_{w, \text{rot}} = \frac{1}{2} I_w \dot{\phi}_w^2. \quad (14)$$

Thus, the total kinetic energy is

$$\begin{aligned} T &= T_{\text{trans}} + T_{\text{corr}} + T_{b, \text{rot}} + T_{w, \text{rot}} \\ &= \frac{1}{2}(m_b + m_w)\dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2}(I_b + m_b \ell^2) \dot{\theta}^2 \\ &\quad + \frac{1}{2} I_w \dot{\phi}_w^2. \end{aligned} \quad (15)$$

Applying the rolling constraint gives

$$\begin{aligned} T &= \frac{1}{2} \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \dot{x}^2 \\ &\quad + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2} (I_b + m_b \ell^2) \dot{\theta}^2. \end{aligned} \quad (16)$$

D. Potential Energy

The gravitational potential energy of the pendulum body is

$$V = m_b g z_b. \quad (17)$$

Substituting (4) gives

$$V = m_b g (z + \ell \cos \theta). \quad (18)$$

Since z is constant, the term $m_b g z$ is constant and does not affect the equations of motion. It can therefore be omitted, yielding

$$V = m_b g \ell \cos \theta. \quad (19)$$

E. Constrained Lagrangian

Using (16) and (19), the constrained Lagrangian is

$$\begin{aligned}\mathcal{L} = & \frac{1}{2} \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \dot{x}^2 \\ & + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ & + \frac{1}{2} (I_b + m_b \ell^2) \dot{\theta}^2 - m_b g \ell \cos \theta.\end{aligned}\quad (20)$$

F. Equations of Motion

The motor applies equal and opposite torques to the wheel and pendulum body. Let M_{mot} denote the motor torque, with positive torque defined in the positive θ direction on the pendulum body.

The generalized motor torque acting on the pendulum is therefore

$$Q_\theta = M_{\text{mot}}. \quad (21)$$

Because the wheel and pendulum body experience equal and opposite motor torques, the corresponding wheel torque is

$$Q_{\phi_w} = -M_{\text{mot}}. \quad (22)$$

After applying the rolling constraint, the motor contributes an effective generalized force in the x direction of

$$Q_x = -\frac{M_{\text{mot}}}{r_w}. \quad (23)$$

The signs in these equations follow the convention that positive θ and positive M_{mot} correspond to clockwise rotation of the pendulum from the upright position.

1) *Translational Equation:* Applying the Euler–Lagrange equation to x gives

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{x}} = & \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \dot{x} \\ & + m_b \ell \dot{\theta} \cos \theta.\end{aligned}\quad (24)$$

Taking the time derivative gives

$$\begin{aligned}\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = & \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \ddot{x} \\ & + m_b \ell \left(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta \right).\end{aligned}\quad (25)$$

Since the Lagrangian does not explicitly depend on x ,

$$\frac{\partial \mathcal{L}}{\partial x} = 0. \quad (26)$$

Therefore, the translational equation of motion is

$$\begin{aligned}\left(m_b + m_w + \frac{I_w}{r_w^2} \right) \ddot{x} \\ + m_b \ell \cos \theta \ddot{\theta} - m_b \ell \sin \theta \dot{\theta}^2 = -\frac{M_{\text{mot}}}{r_w}.\end{aligned}\quad (27)$$

2) *Pendulum Angular Equation:* Applying the Euler–Lagrange equation to θ gives

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = & m_b \ell \dot{x} \cos \theta \\ & + (I_b + m_b \ell^2) \dot{\theta}.\end{aligned}\quad (28)$$

Taking the time derivative gives

$$\begin{aligned}\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = & m_b \ell \left(\ddot{x} \cos \theta - \dot{x} \dot{\theta} \sin \theta \right) \\ & + (I_b + m_b \ell^2) \ddot{\theta}.\end{aligned}\quad (29)$$

The partial derivative with respect to θ is

$$\frac{\partial \mathcal{L}}{\partial \theta} = -m_b \ell \dot{x} \dot{\theta} \sin \theta + m_b g \ell \sin \theta. \quad (30)$$

Therefore, the angular equation of motion is

$$\begin{aligned}(I_b + m_b \ell^2) \ddot{\theta} + m_b \ell \cos \theta \ddot{x} \\ - m_b g \ell \sin \theta = M_{\text{mot}}.\end{aligned}\quad (31)$$

The first term represents the rotational inertia of the pendulum body, the second represents translational–rotational coupling, and the third represents the gravitational torque.

G. Symbol and Coordinate Definitions

TABLE I
SYMBOLS AND COORDINATE DEFINITIONS.

Symbol	Definition
$\mathcal{F}_I$	Inertial Cartesian coordinate frame.
x	Horizontal position of the wheel axle, positive to the right.
z	Vertical position of the wheel axle, positive upward; treated as constant.
θ	Pendulum-body angle measured clockwise from the positive z axis; $\theta = 0$ corresponds to the upright configuration.
ϕ_w	Angular position of the wheel.
ω_w	Wheel angular velocity, $\omega_w = \dot{\phi}_w$.
$\mathbf{r}_w$	Position vector of the wheel axle.
$\mathbf{r}_b$	Position vector of the pendulum-body center of mass.
x_b, z_b	Horizontal and vertical coordinates of the pendulum-body center of mass.
ℓ	Distance from the wheel axle to the pendulum-body center of mass.
r_w	Wheel radius.
m_b	Mass of the pendulum body.
m_w	Mass of the wheel and axle assembly.
I_b	Mass moment of inertia of the pendulum body about its center of mass.
I_w	Mass moment of inertia of the wheel and axle assembly about the wheel axis.
g	Gravitational acceleration.
T	Total kinetic energy of the system.
V	Gravitational potential energy of the system.
$\mathcal{L}$	Lagrangian, $\mathcal{L} = T - V$.
q_i	i th generalized coordinate.
Q_i	Generalized nonconservative force associated with q_i .
M_{mot}	Motor torque applied between the wheel and pendulum body.
C	Constant determined by the initial rolling configuration.
t	Time.